
Anmoji: A Novel Body-Controlled Massage Robot

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Abstract

Limb motion controlled robots have been developed and applied in many fields that affect every aspect of people's life. This paper presents a wearable massage robot named Anmoji, which is controlled by motions of a user's head, shoulders, and ankles. Anmoji is controlled in real time and able to provide functions including head massage, neck massage, back massage, leg massage, foot massage and multi-functional holder. Anmoji contributes a novel combination between massage and limb controlled robot, enabling a host of assistive interactions with the massage actions.

Author Keywords

Intelligent hardware; Massage robot; Wearable device; Body control.

CCS Concepts

- Human-centered computing~Interaction devices

Introduction

Assistive robots are becoming very popular nowadays. A group at the Beijing University of Posts and Telecommunications has created a therapeutic massage robot that can simultaneously massage the shoulders, neck, waist, and chest [1]. The therapeutic robot is a wearable massage device just designed to massage the upper body, and its action of shoulders, neck, waist and

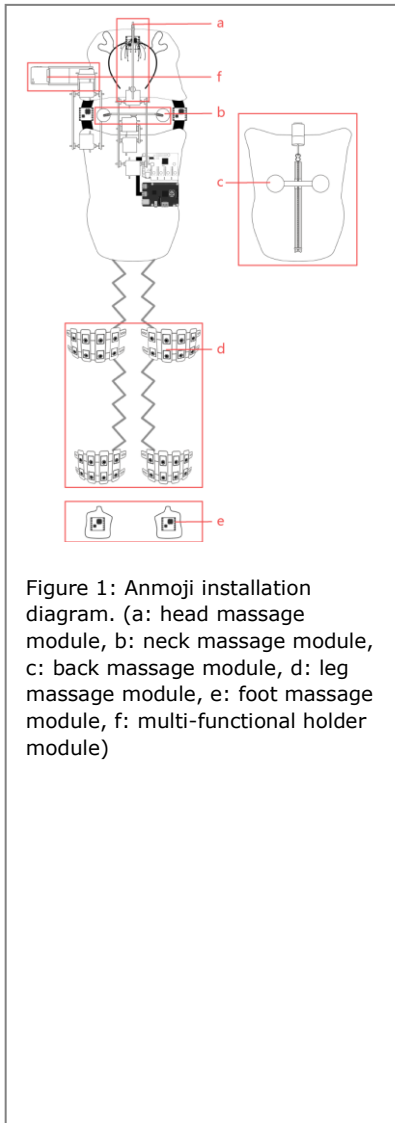


Figure 1: Anmoji installation diagram. (a: head massage module, b: neck massage module, c: back massage module, d: leg massage module, e: foot massage module, f: multi-functional holder module)

chest massage is controlled without input from the limbs. Smart wearable devices that are controlled by limb motion have also increased in recent years. A group of Japanese engineers has designed and built “MetaLimbs” an extra set of robotic arms and hands that are controlled with the feet. This device gives users two additional robotic arms that are manipulated by leg movements. Limb control is achieved using two tracking systems: global motion tracking of the legs using an optical tracker and local motion tracking for manipulating the arms using sock-like devices [2].

We designed a limb-controlled wearable device called Anmoji, which differs from the two examples above in its control system. The user’s limbs are used to control the device by moving the head, shoulders, and ankles. Control inputs from these limbs allow hands-free control of five massage modules and one multi-functional holder module when both of users’ hand are occupied (e.g., working or holding bags).

Anmoji

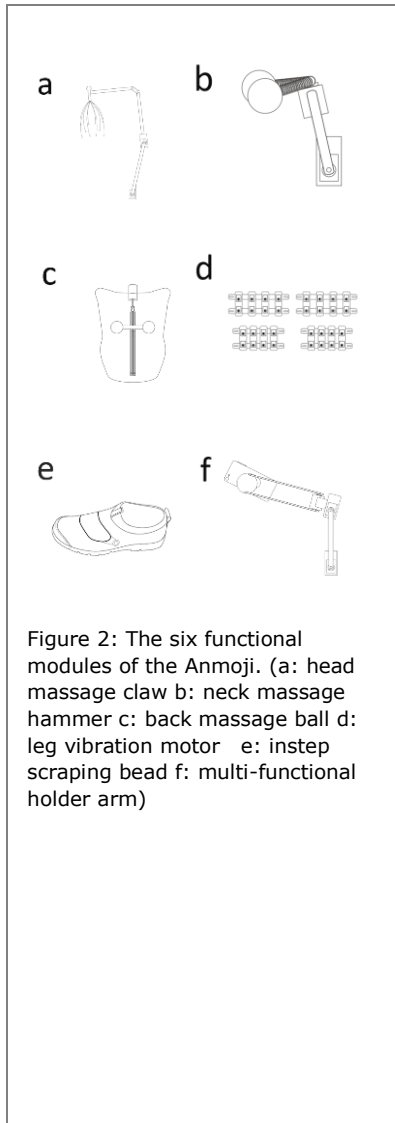
Anmoji is a wearable massage robot that is controlled by the wearer’s head, shoulders, and ankles. Using the Makeblock mBot Ultimate 2.0 (10-in-1 Robot Kit) hardware and built-in sensors as a starting point, we designed an intelligent control unit that processes inputs from the head, shoulders, and ankles. The robot’s six functional modules are head massage module, neck massage module, back massage module, leg massage module, instep massage module, and a multi-functional holder module.

The head massage module is placed above the back of a fixed vest (Fig. 1-a). The neck massage module is placed behind the back of the fixed vest (Fig. 1-b). The back massage module is fixed in the centre of the back

of the fixed vest (Fig. 1-c). The leg massage module is designed like a buttoned garter, tightly wrapped around the user’s thighs and calves (Fig. 1-d). The foot massage module wraps through a shoe cover, and is fixed on the instep (Fig. 1-e). The multi-functional holder module placed on the upper left side of the back of the fixed vest (Fig. 1-f). As shown in Fig. 2, the head massage device includes a mechanical arm and a massage claw (Fig. 2-a), and the elastic massage claw is used for grasping massage. The neck massage device includes a mechanical arm and a massage pounder (Fig. 2-b), mainly used for assault massage. The back massage device includes a fishbone skeleton track and a massage ball (Fig. 2-c), which is used for ball-roller massage. The leg massage module is fastened with a button-down strap and has 16 miniature vibrating motors on each leg. This part tightly wraps around the user’s thigh and calf, and can be adjusted according to the size of the user’s leg (Fig. 2-d) to give vibrating massage. The instep massage device is fixed to the instep according to the design of the user’s shoe cover, and the guide rails fixed on both sides of the sole drives the balls back and forth on an elastic rope, applying a scraping massage (Fig. 2-e). The multi-function holder device includes a mechanical arm and holder (Fig. 2-f), which can hold a phone when the hands are occupied.

Design of interaction

User interaction with Anmoji is mediated through the user’s head, shoulder and ankle movement. Functions are adjusted with shoulder movements. During the massage process, the left shoulder is shrugged quickly, and the corresponding function is determined (Fig.3). The user can shrug the right shoulder once to cancel the corresponding function. In addition, the user can shrug both shoulders at the same time to lock the



corresponding function. This interface allows function switching and adjustment.

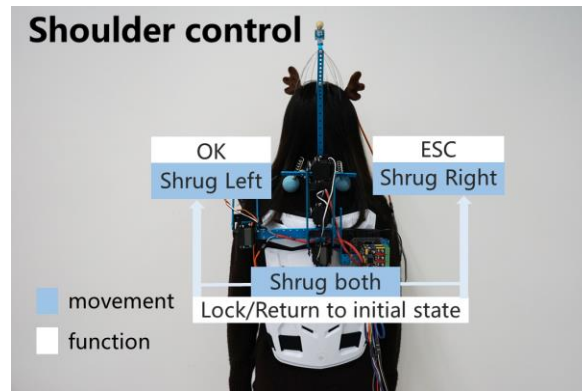


Figure 3: Interaction via shoulder movement.

The user can lean their head forward, backward and look left or right to choose the function (Fig.4). The user triggers the neck massage module with a quick tilt backward and triggers the multi-functional holder module with a quick tilt to the left. A quick tilt to the right triggers the head massage module. When engaging in a function module, the movement and posture of the head will adjust the massage effects.

The user can lean their head forward, backward, left or right to adjust the position of the massage module and control the massage intensity. Taking the head massager as an example, when the head leans forward, the head massage claw gradually approaches the head; when the head leans backward, the claw moves away from the head gradually. If the user wants to change the intensity of massage, the right shoulder is shrugged to return to the previous frequency selection. When the user quickly tilts the head left once, and shrugs the left shoulder, the massage module will decrease its

frequency. The same motion to the right increases the frequency.

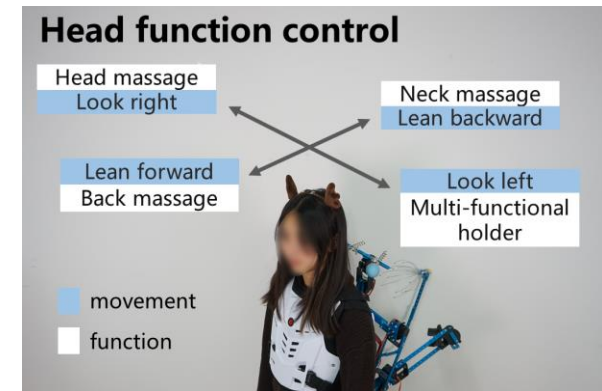


Figure 4: Interaction method for head function control. (a: head massage b: neck massage c: back massage d: multi-functional holder)

When the user steps their ankles up and down, the lower body massage-function modules will respond according to the order of steps. Taking four steps as a cycle, the first step will trigger the whole upper body function, the second will trigger the instep scraping massage module, the third triggers calf massage module, and the forth triggers the thigh massage module (Fig. 5). If the user wants to experience multiple lower-body massage functions at the same time, the left shoulder is shrugged to select the function on after the corresponding angle movement. Then, the shoulders are shrugged once quickly to lock this function. After the current function is locked, the user steps the ankles to select the next function. Using this interaction scheme, the user can operate multiple functions simultaneously.

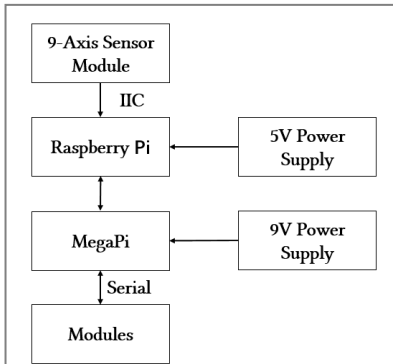


Figure 6: System flow chart of the Anmoji robot.

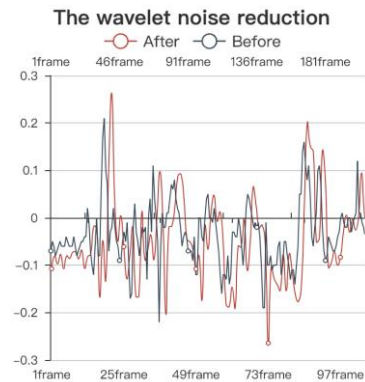


Figure 7: Data (Before) and the filtered noise-reduced data (After).



Figure 5: Interaction scheme for ankle control.

System Development

The Anmoji system architecture includes two main components: a motion capture system and a massage system. When the user move body, the 9-axis sensor module in the motion capture system is first engaged. The motion state of the head, shoulders, and ankles is detected, and this data is passed to Raspberry Pi, where the data will be processed and analyzed using python scripts running on the Raspbian operating system. Then, mechanical control is applied through the MegaPi, to control the massage and bracket modules. The system hardware consists of five 9-axis sensor modules, which are fixed on the hair bands, shoulder armour on both sides, and at the heels of the shoes. The 9-axis sensor module detects attitude and movement, and generate nine data points of Acceleration, Angular velocity, and Angle in real time. Finally, the Raspberry Pi receives and classifies the nine kinds of data and performs time series analysis. The theory. The discrete wavelet transform (dB4) is selected to perform 2-level wavelet decomposition, and filtered noise-reduced data is finally obtained.

Comparing the original data (Before) and the filtered noise-reduced data (After) (Fig.7), we see that the filtering noise reduction method reduces the number of low amplitude signals and enhances the smoothness of the signal.

Conclusion & Future work

We present a wearable massage robot that is controlled by specific body movements. The device recognizes the motion of the user's head, shoulders and ankles in real time with gyroscope modules, and controls the six functional modules accordingly to provide the massage and holder functions. The main contribution of this work is the combination of massage and limb control in a totally hands-free manner. In the future, we plan to improve the ductility of the device, develop new forms of massage and optimize the structure of the device to improve its comfort level.

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